

CLINISCOPE

Healthcare Record Contradiction Detector
Find the conflict. Trace the evidence. Keep humans in control.

TEAM [Add official team name]

Team leader: [Add team leader]

Team members: [Add member names]

Problem Statement

PS-11R3: Healthcare Record Contradiction Detector. The framing below is the team's interpretation.

01 THE PROBLEM

Patient information is spread across digital and scanned documents. Allergies, doses, diagnoses and lab results can differ, and spotting that by hand is slow.

02 TARGET USERS

Intended users: clinicians reviewing records, medical-records (HIM) teams, clinical quality and patient-safety reviewers, and authorized staff reconciling records.

03 CURRENT CHALLENGES

- Fragmented: statements sit in separate files
- Context: dates, negation, dose changes
- Traceability: each conflict needs its source

04 THE GAP / IMPACT

A warning is not enough. Reviewers need the original evidence, its context (dates, history) and a controlled way to record their decision.

ONE-LINE PROBLEM: How can reviewers find and investigate conflicts across medical records with clear source evidence and human oversight?

Proposed Solution

An evidence-first review prototype: documents in, traceable findings out, decisions made by people.

SOLUTION OVERVIEW

CLINISCOPE reads PDF, scanned, image, TXT and DOCX records (OCR for scans), finds clinical statements with rules, compares them across a case and flags potential conflicts with verified source quotes for human review.

UNIQUE VALUE / USP

Every finding links to an exact source quote, with OCR confidence shown. AI suggestions must verify too, and people make the final call.

01 DOCUMENT INGESTION

PDF, scans (on-device OCR), images, TXT and DOCX, read in the browser.

02 EVIDENCE DETECTION

Rules flag conflicts with quotes; optional AI ideas are quote-checked.

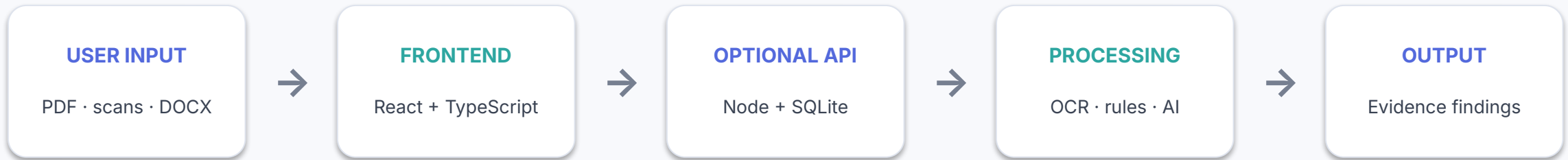
03 REVIEW & AUDIT

Reasons required, audit log, optional shared cases with roles.

EXPECTED BENEFIT: Conflicts become easier to find and investigate, with evidence and human judgment kept.

Technical Approach & Architecture

Browser app with on-device OCR. An optional Node API adds shared cases and AI (not yet publicly hosted).



SYSTEM COMPONENTS

OCR (Tesseract.js), pdf.js, mammoth → statement rules
→ detection → optional AI → quote check → review,
stored in IndexedDB or SQLite.

DATA FLOW & VALIDATION

Uploads are type- and size-checked. Any finding, AI
included, is kept only if its quotes match the source;
low OCR confidence lowers evidence.

ARCHITECTURE NOTE: Works fully in the browser; the server and AI are optional, separately tested add-ons.

Technology Stack

Technologies verified in package.json and the CI workflow, and the role each one plays.

FRONTEND

React 18, TypeScript, Vite and Tailwind CSS for the review UI.

BACKEND

Optional Node 22 API for accounts, roles and case sharing.

DATABASE

IndexedDB (Dexie) in the browser; SQLite on the server.

AI / CORE LOGIC

Deterministic rules; optional Anthropic API, quote-verified.

INTEGRATIONS

Tesseract.js OCR, pdf.js and mammoth run on-device.

DEPLOYMENT & TOOLS

GitHub Actions CI, GitHub Pages; Docker/Render for the API.

WHY THIS STACK? The browser app works on its own, so the demo needs no server. The API and AI layer are optional and add sharing and AI only where configured.

Feasibility & Viability

A working, tested prototype today, with a staged and cautious path toward real-world use.

TECHNICAL FEASIBILITY

- Required tools and skills: React, TypeScript, Node, SQLite, Tesseract.js, Vitest, Playwright.
- Data / infrastructure: synthetic records only. Frontend is live; the API is not hosted yet.
- Prototype status: 68 unit and 12 end-to-end tests pass. AI not yet tried on a live model.
- Risk + mitigation: OCR or AI errors → quote checks, confidence labels, human review.

BUSINESS / OPERATIONAL VIABILITY

- Target users and value: authorized reviewers who need evidence, not just alerts.
- Adoption plan: synthetic prototype → test set → clinician feedback → approved pilot.
- Cost and maintenance: API hosting, storage, AI usage per analysis, testing.
- Revenue (future only): institutional licensing or integration. No partners today.

KEY TAKEAWAY: A working prototype with a practical, validation-first path to evidence-based review.

Sustainability, Scalability & Impact

Scalability items are planned next steps. Impact is a goal to be measured in evaluation.

LONG-TERM SUSTAINABILITY

Modular engine, regression tests in CI, versioned migrations, and low-confidence OCR flagged, never guessed.

SCALABILITY

Planned: host the API, test AI on a live model, more formats and vocabulary, record-system integration.

EXPECTED IMPACT

Aims to reduce manual searching, make investigations traceable and record review decisions consistently.

FUTURE ROADMAP



01 Prototype



02 Pilot testing



03 Improve



04 Deploy & scale

SUCCESS METRIC: Precision/recall, OCR error rate on labelled scans, % findings with evidence.

Conclusion

CLINISCOPE helps reviewers find potential conflicts across digital and scanned healthcare records, trace each one to source evidence, and record decisions with human oversight.

Find the conflict. Trace the evidence. Keep humans in control.

THANK YOU

[Add team email] • github.com/anirudhleetcodes-max/cliniscope-healthcare-contradiction-detector
Live demo: anirudhleetcodes-max.github.io/cliniscope-healthcare-contradiction-detector (scan QR)

